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Notation Note. In the appendix, we use Speaker A/B interchangeably with Speaker 1/2 from the main text, where Speaker B (= Speaker 2) denotes the original poster and Speaker A (= Speaker 1) denotes the other party. Similarly, $B \rightarrow A$ corresponds to $2 \rightarrow 1$.

A Unified Benchmark Data

A.1 Collection and Preprocessing

We construct a unified benchmark from nine existing datasets (Table 7). Each example is converted to a common schema: a natural language question, one correct answer, one incorrect answer, and dataset-specific metadata (e.g., domain, difficulty, bias condition). For eight datasets, we reserve a held-out evaluation split and use remaining instances for training; TruthfulQA is used exclusively for evaluation. Table 8 summarizes split sizes; Table 9 provides examples.

Adversarial Factual QA. **TruthfulQA** (Lin et al., 2022) contains 790 adversarial questions across 38 categories designed to elicit common misconceptions. We use the multiple-choice variant, treating the gold answer as correct and remaining options as incorrect. We use TruthfulQA to measure truthfulness under adversarial prompting, and because of its status as a widely used benchmark, we reserve the entire dataset for evaluation.

Hallucination and Plausibility. **HaluEval** (Li et al., 2023) pairs questions with gold answers and hallucinated model outputs; we treat hallucinated variants as incorrect. **PlausibleQA** (Mozafari et al., 2025) provides questions with plausibility-scored candidates; we use high-plausibility but incorrect candidates as challenging distractors. Together, these datasets probe whether models resist attractive but incorrect completions.

Bias and Fairness. **BBQ** (Parrish et al., 2022) targets nine social dimensions (race, gender, religion, etc.) with ambiguous and disambiguated question variants. We concatenate context and question into a single prompt and treat the statistically correct or “unknown” option as correct per the authors’ labels. We use BBQ to analyze how conversational framing interacts with social bias under controlled conditions.

Medical QA. **AMQA** (Xiao et al., 2025) draws from USMLE questions, constructing clinically equivalent variants differing only in sensitive attributes (race, sex, socioeconomic status). We preserve the original multiple-choice structure and bias metadata. We use AMQA to examine both clinical accuracy and disparities across patient groups.

Social Reasoning. **SocialIQA** (Sap et al., 2019) contains multiple-choice questions about everyday

Category	Dataset	Domain	Source Type	Evaluation Focus
<i>Factual QA</i>	TruthfulQA (Lin et al., 2022)	General	Adversarial QA	Resistance to imitative falsehoods and common misconceptions
	HaluEval (Li et al., 2023)	General	Hallucination	Detecting machine-generated hallucinations
	PlausibleQA (Mozafari et al., 2025)	General	QA pairs	Distinguishing truth from plausible distractors
<i>Social Reasoning</i>	BBQ (Parrish et al., 2022)	Social Bias	Template-based	Bias mitigation across 9 demographic dimensions
	SocialIQA (Sap et al., 2019)	Commonsense	Crowdsourced	Reasoning about social motivations and reactions
<i>Specialized</i>	AMQA (Xiao et al., 2025)	Medical	Exam (USMLE)	Clinical accuracy and demographic bias
	HARP (Yue et al., 2024)	Math	Competitions	High-precision reasoning (AMC, AIME)
	GPQA (Rein et al., 2024)	Science	Expert-written	Graduate-level scientific reasoning
<i>Subjective</i>	AdvisorQA (Kim et al., 2025)	Advice	Reddit (r/LPT)	Community preference for helpfulness/safety

Table 7: **Unified Benchmark: Dataset Descriptions.** Each instance consists of a question, one correct answer, and one incorrect answer, yielding four experimental cases (C_1 -True, C_1 -False, C_2 -Correct, C_2 -Incorrect).

social situations, asking about motivations, reactions, or next actions. We concatenate context and question and designate the gold answer as correct. We use SocialIQA to evaluate non-factual, socially grounded reasoning central to conversational agents.

Math and Science. **HARP** (Yue et al., 2024) collects 4,110 multiple-choice problems from U.S. math competitions (AJHSME, AMC, AIME, USA(J)MO) spanning multiple difficulty levels. **GPQA** (Rein et al., 2024) provides 448 expert-written graduate-level questions in biology, physics, and chemistry. We preserve difficulty levels and subdomains in metadata. We use HARP and GPQA to stress-test models’ ability to make precise judgments in mathematically and scientifically demanding settings.

Subjective Advice. **AdvisorQA** (Kim et al., 2025) contains advice-seeking posts from *r/LifeProTips* and *r/UnethicalLifeProTips* with ranked human answers. We use the top-ranked answer as correct and retain upvote signals and safe/unsafe labels in metadata. We use AdvisorQA as a non-factual but high-stakes testbed where correctness is defined by helpfulness and safety rather than verifiable truth.

A.2 Train–Test Splits and Stratified Sampling

For each dataset except TruthfulQA, we construct held-out evaluation splits (Table 8). When meta-

data defines meaningful groups, we use stratified sampling: **AdvisorQA** by safety label, **BBQ** by context condition, **GPQA** by scientific domain, **HARP** by difficulty level, and **SocialIQA** by prompt dimension. For **AMQA**, **HaluEval**, and **PlausibleQA**, we sample uniformly at random (seed 42) because available metadata is too fine-grained for stable strata.

TruthfulQA is reserved entirely for evaluation to ensure truthfulness results are measured on a completely held-out standard benchmark. All splits use unique id fields to prevent train–test leakage.

Dataset	Total	Train	Test	Sampling	Balancing Strategy
r/AIO [†]	280	—	280	Full	—
TruthfulQA [†]	790	—	790	Full	—
AdvisorQA	1,000	700	300	Balanced	150 safe + 150 unsafe
AMQA	801	561	240	Random	—
BBQ	58,492	58,192	300	Balanced	150 ambiguous + 150 disambiguated
GPQA	448	314	134	Balanced	~45 per domain (3 total)
HaluEval	10,000	9,700	300	Random	—
HARP	4,110	3,810	300	Balanced	75 per difficulty level (1–4)
PlausibleQA	10,000	9,700	300	Random	—
SocialIQA	2,224	1,924	300	Balanced	~33 per prompt dimension
Total	87,865	85,001	2,864	—	—

Table 8: **Dataset Splits and Sampling Strategy.** [†]TruthfulQA and r/AIO are evaluated in full and excluded from training. For balanced datasets, samples are distributed evenly across categories; random datasets use simple random sampling. Train set is reserved for SFT/DPO mitigation experiments (§4.1). All sampling uses seed 42.